DA6701

Assignment IV: Portfolio Replication

Team 14

April 5, 2026

1 Problem Overview

The objective of this assignment is to construct a sparse portfolio that closely tracks the return profile of the S&P 500 index while using only a limited subset of its constituents ($k \leq 50$).

This problem introduces a fundamental trade-off between *sparsity* and *tracking accuracy*, and can be formally written as:

$$\min_{w} \ \mathrm{Var}(R_p - R_b)$$

subject to:

$$||w||_0 \leq k, \quad \sum w_i = 1$$

The presence of the ℓ_0 constraint renders the problem NP-hard, making classical optimization techniques insufficient. We therefore employ a combination of regularization-based and heuristic methods.

2 Data and Setup

We use daily OHLCV data for S&P 500 constituents from Jan 2020 to Dec 2025.

The dataset is split temporally to ensure realistic evaluation:

- Training: Jan 2020 – June 2025
- Holdout: July 2025 – Dec 2025

Daily returns are computed as:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

All computations strictly avoid look-ahead bias.

3 Methodology

The objective of this project is to construct a sparse portfolio that replicates the daily return profile of the S&P 500 using a limited subset of its constituents. The dataset spans January 2020 to February 2026, with January 2020 to June 2025 used for model development and July 2025 to December 2025 reserved as a holdout test set. The goal is to minimize tracking error while enforcing a cardinality constraint on the number of selected stocks.

Let $R_p(t)$ and $R_b(t)$ denote the portfolio and benchmark returns, respectively. The tracking error is defined as:

$$\mathrm{TE} = \sqrt{\mathrm{Var}(R_p - R_b)}. \tag{1}$$

Each method produces a sparse portfolio, which is evaluated using tracking error, information ratio, and sector allocation on both in-sample and holdout data.

3.1 LASSO-Based Portfolio Replication

LASSO regression is used to construct a sparse tracking portfolio by selecting a subset of stocks while estimating their weights.

Given the return matrix X and benchmark returns y, the optimization problem is:

$$\min_{w} \ \|y - Xw\|_2^2 + \lambda \|w\|_1 \tag{2}$$

where λ controls sparsity.

The model is trained over a range of λ values, producing portfolios with different subset sizes k. For each configuration, weights are normalized to form a fully invested portfolio.

The optimal model is selected based on minimum holdout tracking error.

3.2 Autoencoder-Based Portfolio Replication

This approach uses a deep autoencoder to learn a low-dimensional representation of stock return dynamics and identify a sparse subset of representative stocks.

Data Processing

Daily returns are computed and aligned with the benchmark. Stocks with insufficient data are removed, and the return matrix $X \in \mathbb{R}^{T \times N}$ is standardized. The data is split into fit, selection, and holdout periods.

Model Architecture

A fully-connected autoencoder is used:

- Encoder: $N \to 256 \to 64 \to d$
- Decoder: $d \to 64 \to 256 \to N$

where d is the latent dimension. The model is trained to minimize reconstruction error:

$$\mathcal{L} = \frac{1}{TN} \|X - \hat{X}\|_F^2 \tag{3}$$

Stock Selection

After training, latent representations are used to score stocks based on:

- reconstruction quality (communality),
- contribution to latent factors,
- a hybrid of both.

For each target subset size k, the top-ranked stocks are selected.

Portfolio Construction

Weights are estimated by regressing benchmark returns on the selected stocks:

$$\min_{w} \|y - X_{\mathcal{S}} w\|_2^2 \tag{4}$$

and normalized to form a fully invested portfolio.

Evaluation

The process is repeated for different values of k to analyze the sparsity–tracking error trade-off. Performance is evaluated using tracking error and information ratio on both selection and holdout datasets.

3.3 Genetic Algorithm-Based Portfolio Replication

A genetic algorithm (GA) is used to solve the sparse portfolio selection problem under a cardinality constraint.

Each candidate portfolio is represented as a binary vector indicating selected stocks, with the number of active positions constrained to k. For each candidate, portfolio weights are estimated via regression on the benchmark returns, and the objective is to minimize tracking error.

The fitness function is defined as:

$$\text{Fitness} = -\text{TE}(R_p - R_b) \tag{5}$$

The algorithm proceeds through iterative evolution:

- initialization of a random population,
- selection of high-fitness portfolios,
- crossover to combine parent solutions,
- mutation to introduce diversity,
- repair to enforce the cardinality constraint.

This process is repeated for multiple values of k, and the best-performing portfolio is selected based on minimum holdout tracking error.

3.4 Greedy-Based Portfolio Replication

A greedy forward-selection approach is used to construct a sparse tracking portfolio by iteratively selecting stocks that improve replication performance.

Starting from an empty set, stocks are added sequentially based on a selection criterion. Three variants are considered:

- **Greedy TE**: selects the stock that minimizes tracking error,
- **Greedy Corr+Div**: balances correlation with the benchmark and diversification,
- **Greedy Beta**: selects stocks to match benchmark beta exposure.

At each step, portfolio weights are re-estimated via regression on the benchmark returns. The process continues until the desired subset size k is reached.

The procedure is repeated for multiple values of k, and the best configuration is selected based on holdout tracking error.

3.5 Clustering-Based Portfolio Replication

A clustering-based approach is used to reduce redundancy in the stock universe by grouping similar stocks and selecting representatives from each cluster.

A feature matrix is constructed using return-based statistics (mean, volatility, skewness, kurtosis, and day-of-week effects) and standardized. Dimensionality reduction is performed using t-SNE to capture nonlinear structure in the data.

Two clustering methods are applied:

- **K-Means**: clusters stocks into k groups, selecting one representative per cluster,
- **Affinity Propagation**: automatically determines the number of clusters and selects exemplars.

Selected stocks are used to construct the portfolio, with weights estimated via regression on benchmark returns. The best configuration is chosen based on holdout tracking error.

4 Results

This section presents the empirical performance of all five portfolio replication approaches. Each method is evaluated using tracking error (TE), information ratio (IR), and sector allocation stability on both in-sample and out-of-sample data.

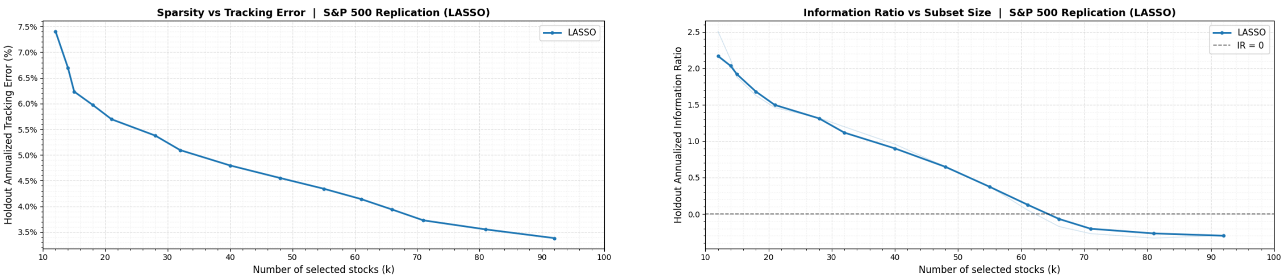
4.1 LASSO-Based Results

Metric	Value
Best α	0.001366
Best k	48
Tracking Error (Holdout)	4.5522%
Information Ratio (Holdout)	0.6496

Table 1: Optimal LASSO configuration

Ticker	Weight	Abs Weight
MSFT	0.1119	0.1119
AAPL	0.1090	0.1090
BRK-B	0.0830	0.0830
APH	0.0699	0.0699
AMZN	0.0666	0.0666
AMP	0.0614	0.0614
BLK	0.0526	0.0526
GOOGL	0.0375	0.0375
TEL	0.0329	0.0329
NVDA	0.0315	0.0315
AVGO	0.0264	0.0264
INTU	0.0256	0.0256
ISRG	0.0194	0.0194
ACN	0.0175	0.0175
CPRT	0.0171	0.0171
TROW	0.0162	0.0162
PAYX	0.0154	0.0154
OKE	0.0152	0.0152
HON	0.0149	0.0149
IQV	0.0143	0.0143

Table 2: Top selected stocks and weights (LASSO)



(a) Tracking Error vs k

(b) Information Ratio vs k

Figure 1: LASSO performance across subset sizes

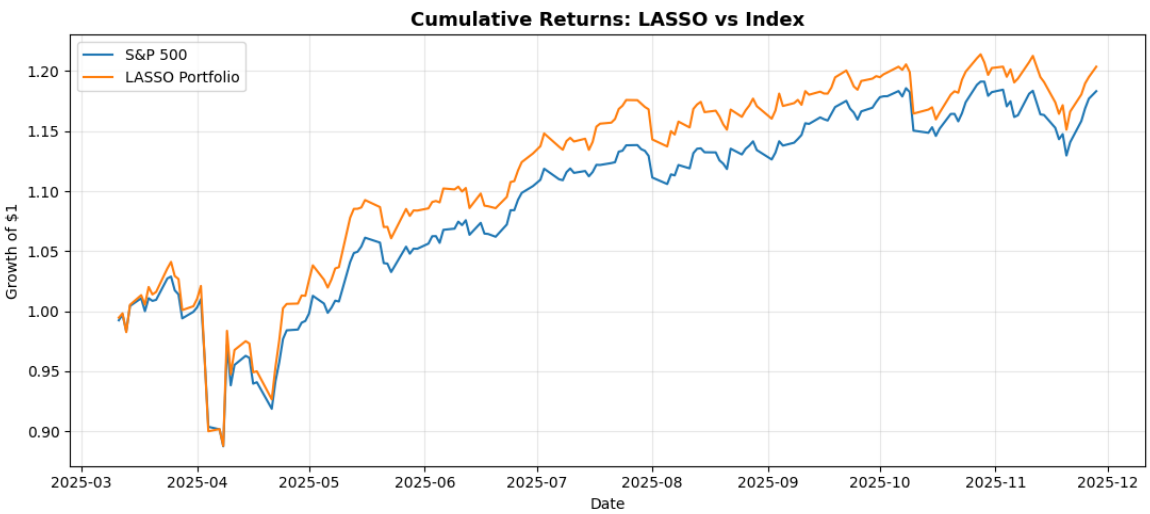


Figure 2: Cumulative returns: LASSO vs S&P 500

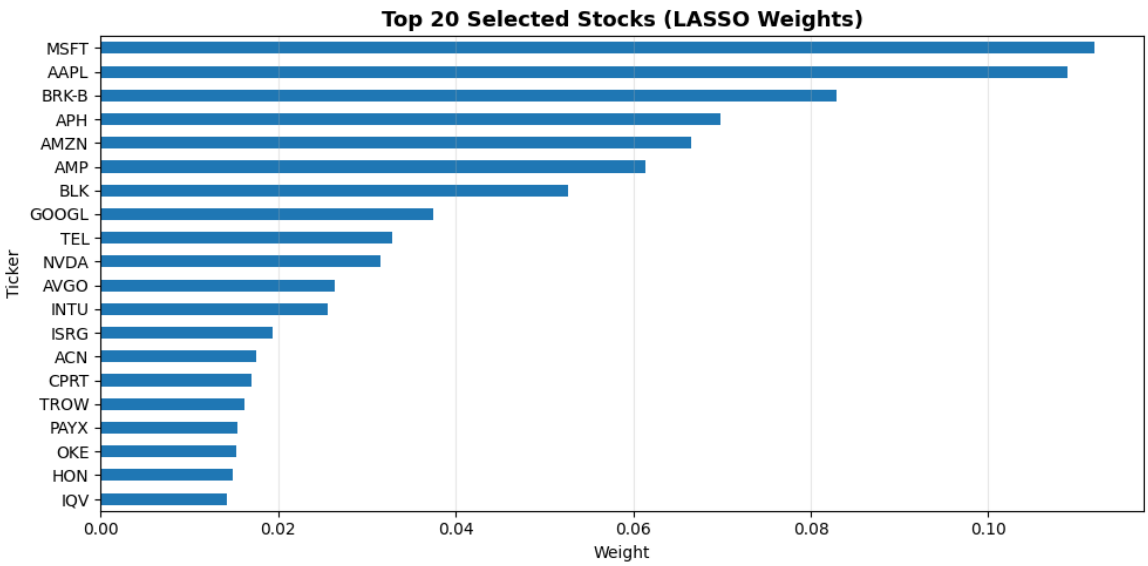


Figure 3: Top 20 selected stocks by weight

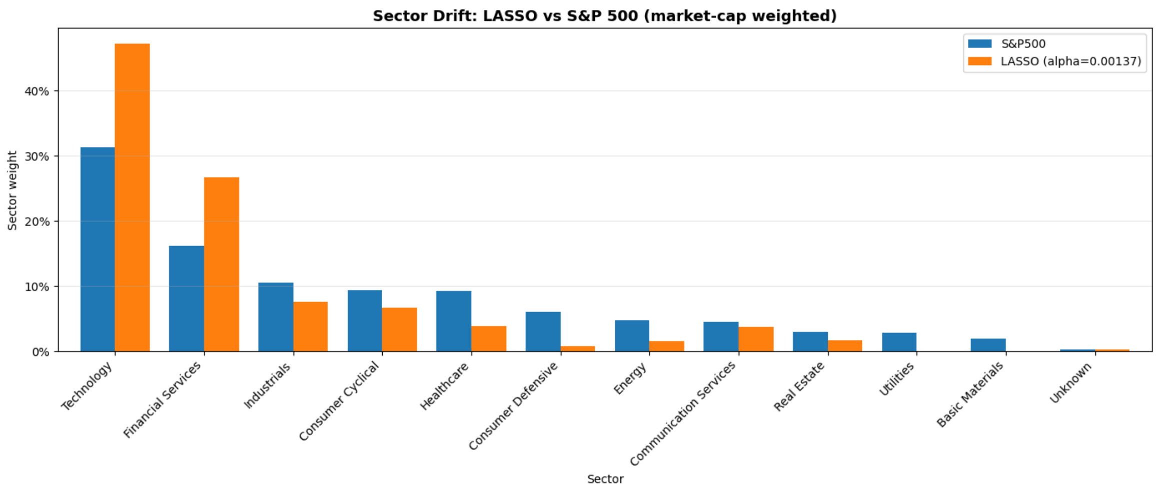


Figure 4: Sector allocation comparison: LASSO vs S&P 500

Sector	S&P 500	LASSO
Technology	0.3135	0.4723
Financial Services	0.1618	0.2673
Industrials	0.1056	0.0759
Consumer Cyclical	0.0933	0.0666
Healthcare	0.0929	0.0386
Consumer Defensive	0.0600	0.0076
Energy	0.0481	0.0152
Communication Services	0.0450	0.0375
Real Estate	0.0289	0.0161
Utilities	0.0285	0.0000
Basic Materials	0.0197	0.0000
Unknown	0.0026	0.0029

Table 3: Sector weight comparison for LASSO

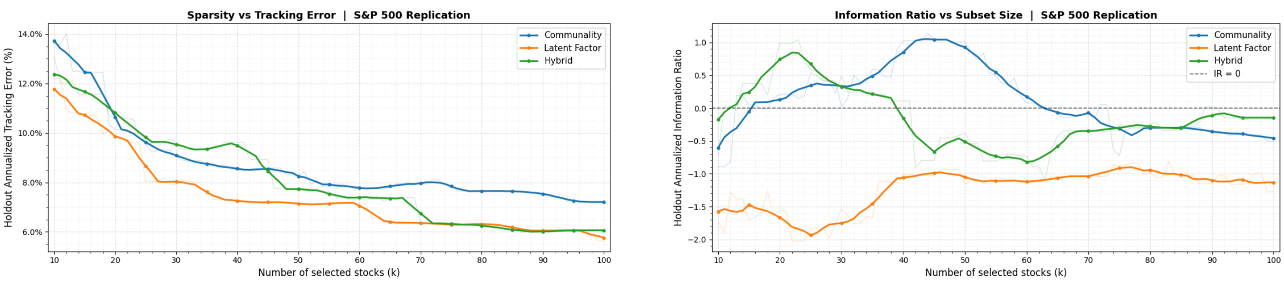
4.2 Autoencoder-Based Results

Metric	Fit	Selection	Holdout
Tracking Error	0.0871	0.0846	**0.0830**
Information Ratio	0.2897	0.5726	**1.0165**
Best k		50	
Best Method		Communality	

Table 4: Best autoencoder configuration (communality-based selection)

Ticker	Weight	Communality Score
CMS	0.1146	0.7359
KEYS	0.0874	0.7550
WEC	0.0844	0.7364
TRMB	0.0718	0.8377
TROW	0.0660	0.7740
TEL	0.0522	0.7687
KLAC	0.0521	0.7631
FTV	0.0521	0.7646
LNT	0.0507	0.7454
MPWR	0.0459	0.7475

Table 5: Top weighted stocks from the best autoencoder portfolio



(a) Tracking Error vs k (b) Information Ratio vs k

Figure 5: Trade-off between sparsity and performance across methods

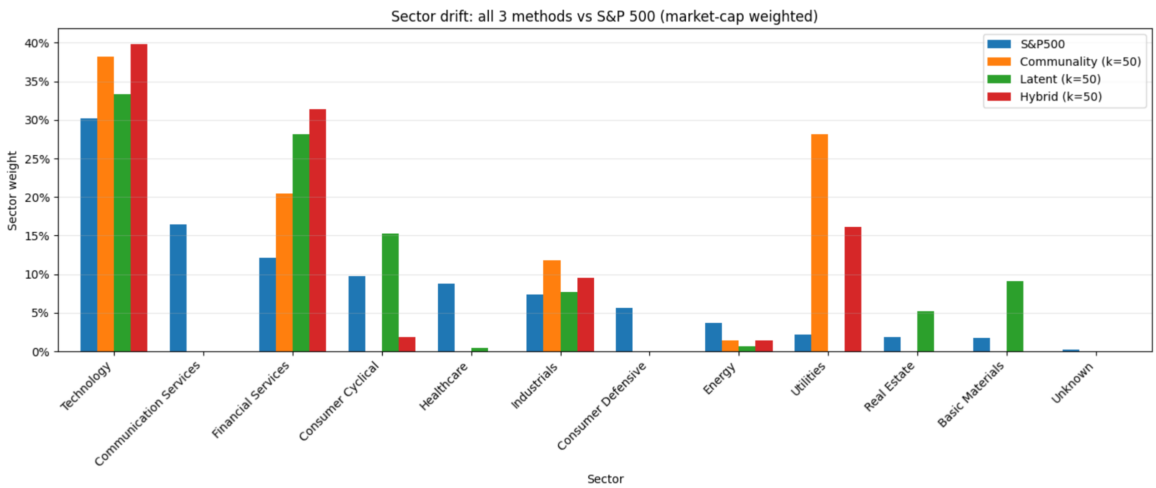


Figure 6: Sector drift comparison: Autoencoder vs S&P 500

Sector	S&P 500	Communality	Latent	Hybrid
Technology	0.3024	0.3820	0.3334	0.3985
Communication Services	0.1645	0.0000	0.0000	0.0000
Financial Services	0.1211	0.2050	0.2814	0.3134
Consumer Cyclical	0.0970	0.0002	0.1530	0.0182
Healthcare	0.0876	0.0000	0.0047	0.0000
Industrials	0.0739	0.1177	0.0769	0.0952
Consumer Defensive	0.0565	0.0000	0.0000	0.0000
Energy	0.0365	0.0142	0.0069	0.0136
Utilities	0.0218	0.2809	0.0000	0.1610
Real Estate	0.0185	0.0000	0.0523	0.0000
Basic Materials	0.0177	0.0000	0.0914	0.0000

Table 6: Sector weight comparison across methods

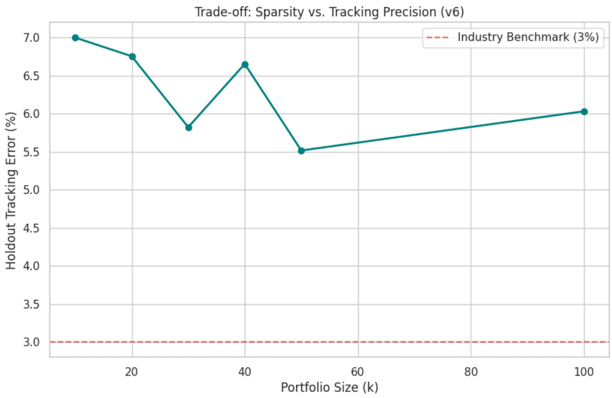
4.3 Genetic Algorithm Results

k	Tracking Error (%)	Beta	R^2	Notes
10	6.9991	0.7869	0.6347	
20	6.7541	0.6672	0.6297	
30	5.8221	0.6878	0.7260	
40	6.6536	0.7042	0.6438	
50	**5.5161**	0.7660	**0.7519**	Best
100	6.0308	0.7484	0.7054	

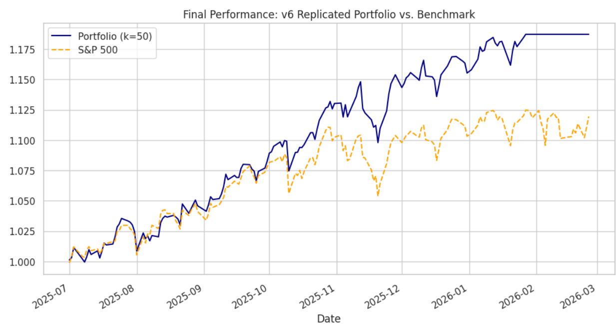
Table 7: Genetic algorithm performance across different subset sizes

Metric	Value
Best k	50
Tracking Error	5.5161%
Portfolio Beta	0.7660
R^2	0.7519
Information Ratio	**1.0494**

Table 8: Final performance metrics for GA portfolio



(a) Sparsity vs Tracking Error



(b) Cumulative Returns vs Benchmark

Figure 7: Performance of GA portfolio across different configurations

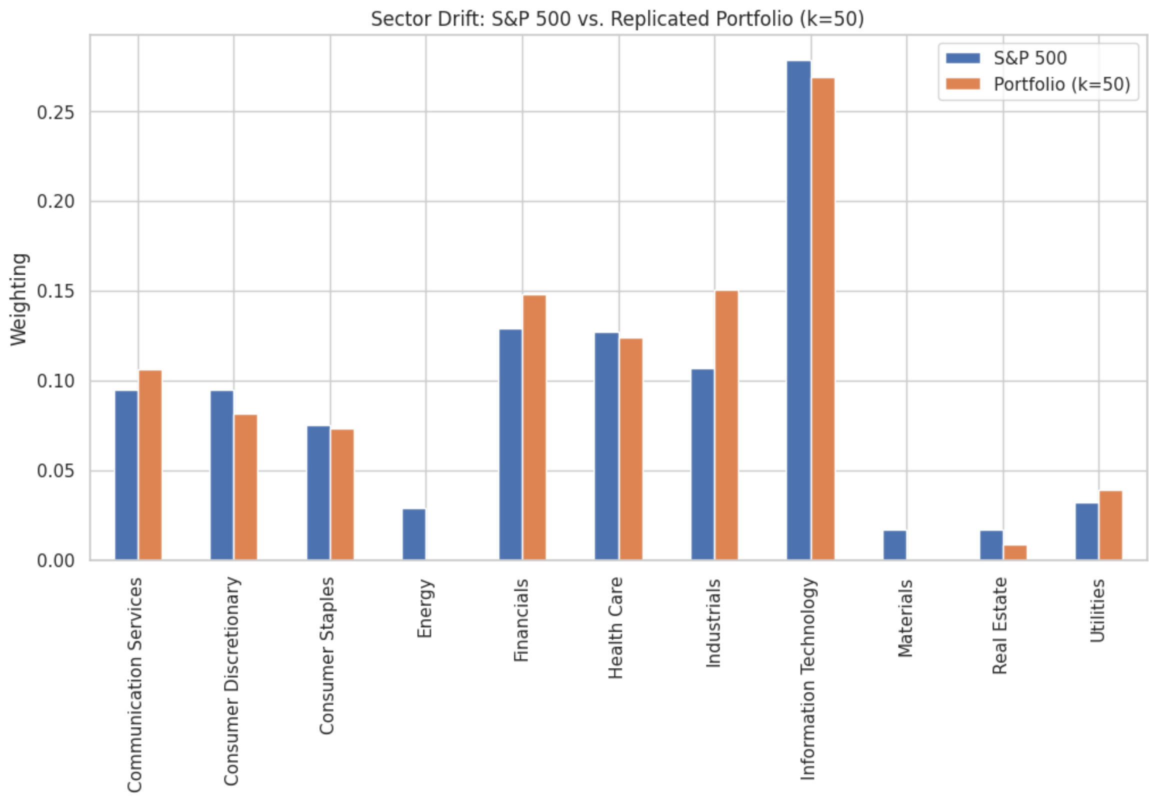


Figure 8: Sector allocation comparison: GA portfolio vs S&P 500

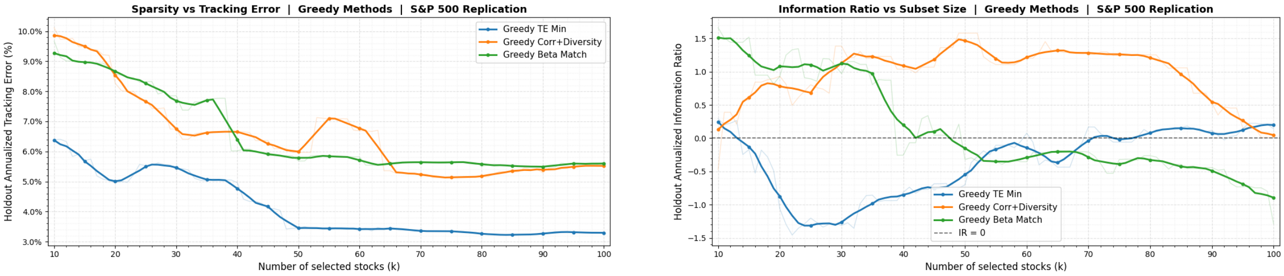
4.4 Greedy Selection Results

Method	Best k	Tracking Error (%)	Information Ratio	Notes
Greedy TE	48	**3.4186**	-0.8580	Lowest TE
Greedy Corr+Div	50	5.9687	**1.5816**	Best IR
Greedy Beta	50	5.6828	-0.3202	Beta-aligned

Table 9: Greedy method performance comparison

Ticker	Weight	Method
BRK-B	0.0696	Greedy TE
MSFT	0.0651	Greedy TE
AAPL	0.0645	Greedy TE
AAPL	0.1275	Corr+Div
CSCO	0.1158	Corr+Div
AMZN	0.0969	Corr+Div
MSFT	0.1897	Beta
JPM	0.0649	Beta
NDAQ	0.0599	Beta

Table 10: Top selected stocks across greedy variants



(a) Tracking Error vs k (b) Information Ratio vs k

Figure 9: Greedy method performance across subset sizes

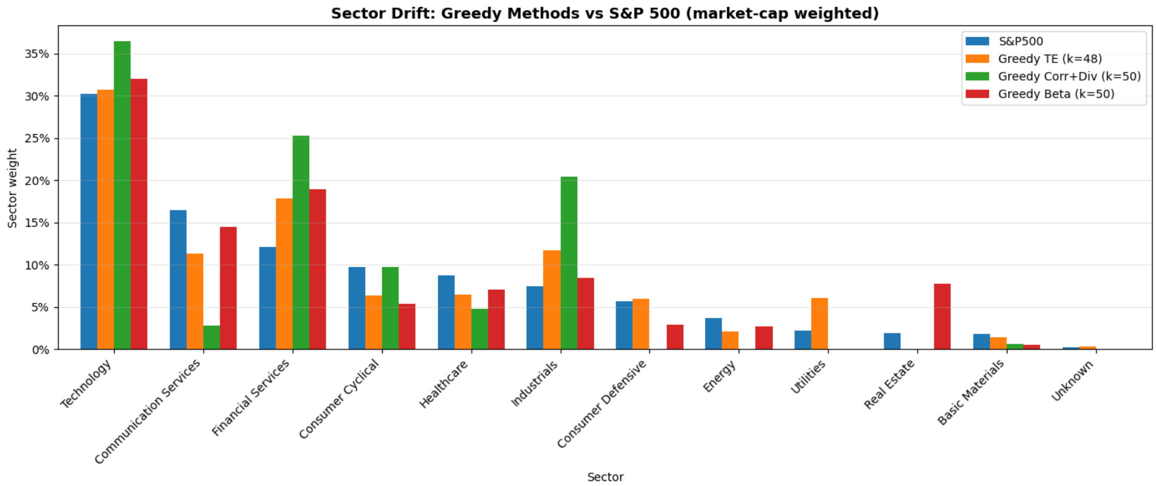


Figure 10: Sector allocation comparison: Greedy portfolios vs S&P 500

Sector	S&P 500	Greedy TE	Corr+Div	Beta
Technology	0.3024	0.3069	0.3649	0.3205
Communication Services	0.1645	0.1127	0.0280	0.1450
Financial Services	0.1211	0.1784	0.2529	0.1894
Consumer Cyclical	0.0970	0.0638	0.0969	0.0537
Healthcare	0.0876	0.0641	0.0473	0.0702
Industrials	0.0739	0.1171	0.2042	0.0840
Consumer Defensive	0.0565	0.0596	0.0000	0.0283
Energy	0.0365	0.0208	0.0000	0.0270
Utilities	0.0218	0.0604	0.0000	0.0000
Real Estate	0.0185	0.0000	0.0000	0.0774
Basic Materials	0.0177	0.0136	0.0058	0.0045

Table 11: Sector weight comparison for greedy methods

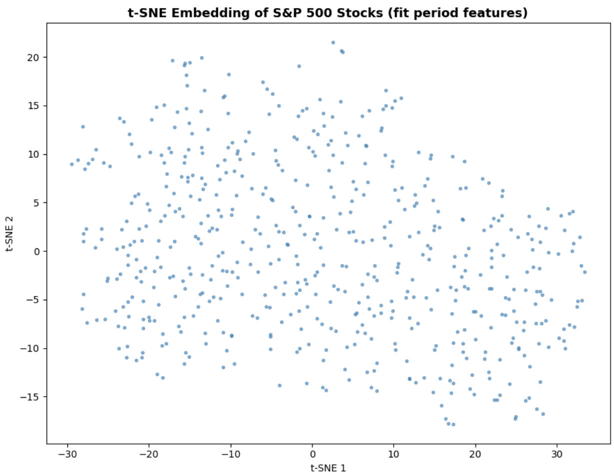
4.5 Clustering-Based Results

Method	k	Tracking Error (%)	Information Ratio	Notes
K-Means	50	**6.4329**	**1.9154**	Best overall
Affinity Propagation	21	8.0800	-1.1931	Auto cluster selection

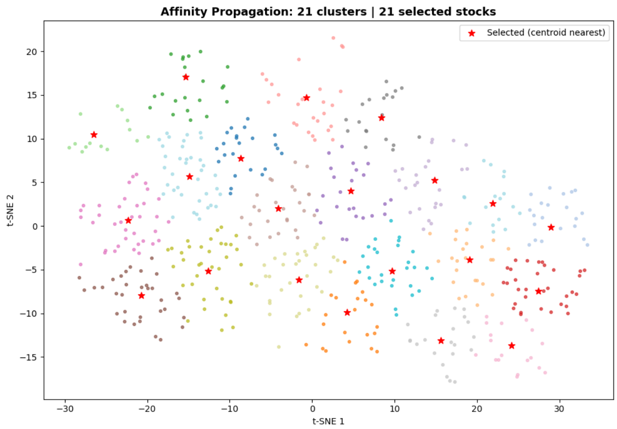
Table 12: Clustering method performance comparison

Ticker	Weight	Method
GOOGL	0.1259	K-Means
ATO	0.0702	K-Means
NVDA	0.0674	K-Means
APH	0.1341	Affinity
KMB	0.1117	Affinity
RSG	0.1012	Affinity

Table 13: Top selected stocks across clustering methods

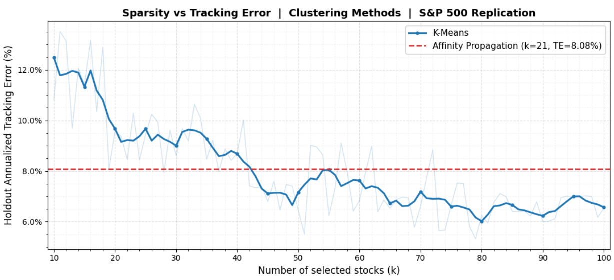


(a) t-SNE embedding

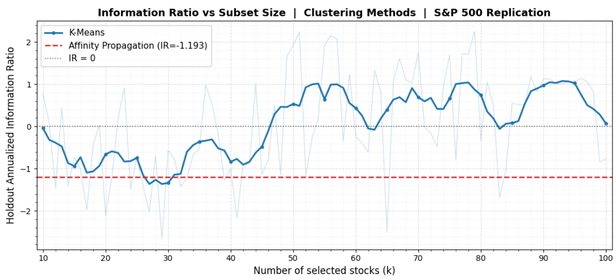


(b) Affinity propagation clusters

Figure 11: Clustering structure of stock universe



(a) Tracking Error vs k



(b) Information Ratio vs k

Figure 12: Clustering method performance across subset sizes

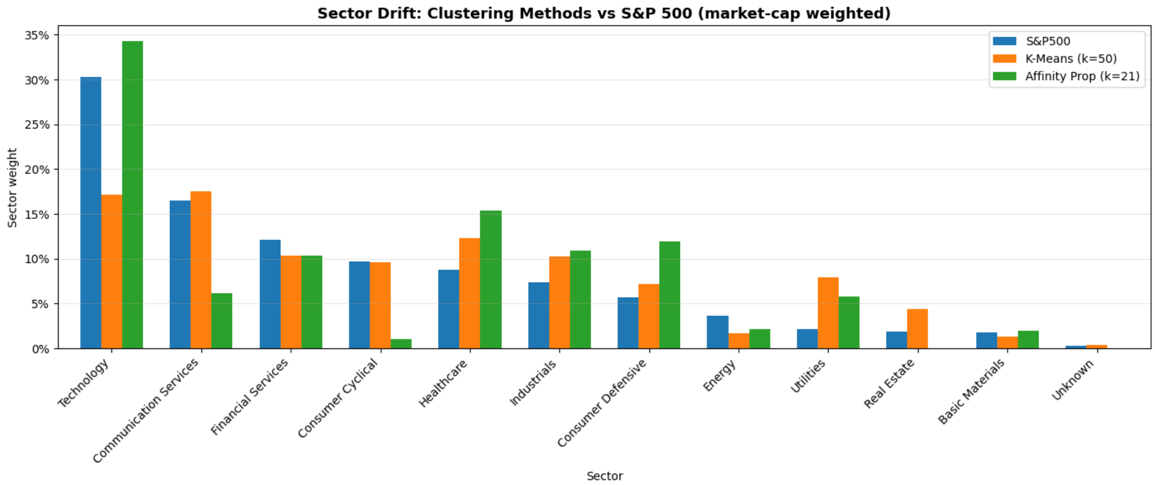


Figure 13: Sector allocation comparison: Clustering vs S&P 500

Sector	S&P 500	K-Means	Affinity
Technology	0.3024	0.1712	0.3432
Communication Services	0.1645	0.1749	0.0614
Financial Services	0.1211	0.1031	0.1037
Consumer Cyclical	0.0970	0.0957	0.0106
Healthcare	0.0876	0.1235	0.1540
Industrials	0.0739	0.1029	0.1092
Consumer Defensive	0.0565	0.0719	0.1194
Energy	0.0365	0.0167	0.0212
Utilities	0.0218	0.0796	0.0575
Real Estate	0.0185	0.0438	0.0000
Basic Materials	0.0177	0.0132	0.0198

Table 14: Sector weight comparison for clustering methods

5 Results and Conclusion

Method	Best k	Tracking Error (%)	Information Ratio	Key Strength
LASSO	48	4.5522	0.6496	Sparse + stable
Autoencoder	50	8.3027	1.0165	Strong generalization
Genetic Algorithm	50	5.5161	1.0494	Robust search
Greedy (TE)	48	**3.4186**	-0.8580	Lowest error
Greedy (Corr+Div)	50	5.9687	**1.5816**	Best IR
Clustering (K-Means)	50	6.4329	1.9154	Highest IR

Table 15: Comparison of all portfolio replication methods

The results highlight a clear trade-off between tracking error and risk-adjusted performance.

Greedy TE achieves the lowest tracking error (3.4186%) but exhibits poor generalization. In contrast, clustering (K-Means) and greedy correlation-diversity methods deliver the highest information ratios, indicating stronger robustness.

LASSO provides the best balance between sparsity and stability, while the genetic algorithm shows consistent performance. The autoencoder captures market structure but is less effective for precise tracking.

All methods show sector bias, typically overweighting technology and financials while underweighting defensive sectors.

Conclusion: No single method dominates. Greedy methods minimize error, clustering-based approaches maximize risk-adjusted returns, and LASSO offers the best overall trade-off.